QUESTION BANK MODULE WISE

MODULE_1

Differential amplifier
- Large Signal Analysis
- Small Signal Analysis
- Input Differential Resistance
- Differential Voltage Gain
- Common Mode Voltage Gain
- CMRR

Concept of current mirror
- *Wilson current mirror*
- *Widlar current mirror*

Basics of OPAMP
- Block Diagram
- Equivalent circuit
- Voltage transfer curve
- Virtual Ground Concept
- Ideal & Practical OPAMP Parameter

MODULE_2

Feedback Amplifier
- *Voltage series feedback*
- *Voltage shunt feedback*

OPAMP Applications (Open Loop)
- *Inverting and Non-inverting amplifiers*
- *Differential Amplifier*

OPAMP Applications (Closed Loop)
- Inverting and Non-inverting amplifiers
- Voltage to current converter
- Current to voltage converter
- Integrator
- Differentiator
- *Precision rectifiers*
- Comparators
- Schmitt Trigger
- Log and antilog amplifier
- *Instrumentation amplifier*

MODULE_3

OPAMP Oscillators
- Phase Shift Oscillator
- Wien-bridge Oscillator
- Triangular Wave generator

OPAMP Multivibrators
Astable and monostable Multivibrators

MODULE_3

Active filters
- 1st Order Active LPF & HPF
- Sallen Key Architecture
- Second order LPF & HPF
- Second order Band pass Filter
- Second order Band reject Filter

MODULE_4

Timer IC 555
- Functional diagram
- Astable and monostable Multivibrators

Phase Locked Loop (PLL)
- Block Diagram
- Free Running Stage
- Lock and capture range
- Functional diagram of IC 565
- *Applications*

Voltage Controlled Oscillator (VCO)
- Functional diagram of IC LM566

MODULE_5

Voltage Regulators
- Fixed and Adjustable
- Functional diagram of IC 723
- Low Voltage Regulator_ IC 723
- High Voltage Regulator_ IC 723
- *Current boosting _ IC 723*
- *Current limiting_ IC 723*

Data Converters

DAC
- Specifications
- Weighted resistor type
- R-2R Ladder type

ADC
- Specifications
- Flash type
- Successive approximation type

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